

DURATION OF ALL SEGMENTS OF USE OF CONSUMER PRODUCTS

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Summary: In this paper, the probability mass function of the mixture of the zero truncated Poisson and the inverse Gaussian distributions is derived. This is applied in modelling the duration of all segments of use of consumer products. In this context, expressions for the survival function, the average duration of use and the termination rates are also derived. The results are applied in studying brand loyalty for a particular brand of toothpaste, using the usage date for four households (Banerjee and Bhattacharya, 1976). It was found that brand loyalty is strongest for household III, followed by households IV and I. It is, however, very weak for household II.

1. Introduction

There are two alternative measures of purchase incidence of frequently purchased low-cost consumer products, namely the number of purchases of the product in a fixed time interval and the waiting time between successive purchases (Banerjee and Bhattacharya, 1976). Purchase incidence models attempt to study the probability distribution of either or both of these measures.

Anscombe (1967), Ehrenberg (1959) and Chatfield et al. (1966) modelled successive purchases of an individual consumer as a Poisson process or, equivalently, using the exponential distribution for interpurchase times. In these studies, heterogeneity of the consumer population was incorporated by

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assuming that the parameter of the exponential distribution follows the gamma distribution. The exponential model was later found to be inappropriate for the obvious reason that the purchase process of an individual consumer is not likely to have the "lack of memory" property. Hermiter (1971) and Chatfield and Goodhardt (1973) considered a one-parameter gamma model for interpurchase time. They observed that the average value of the estimated shape parameter of the gamma distribution is close to the integer 2 and therefore assumed the fixed value 2 for the shape parameter in fitting the gamma distribution to the purchase data of a number of individual households. Thus, only the scale parameter is allowed to vary in their model. Their model is, however, defective since empirical evidence shows a wide variability in the shape parameter (Chatfield and Goodhardt, 1973). The reason for fixing the shape parameter at the observed average value may be as a result of the mathematical difficulty associated with a Bayesian analysis of the two-parameter gamma family.

Chatfield et al. (1966) proposed the negative binomial distribution (NBD) for modelling household purchases of ordinary packaged consumer goods. As pointed out by Chatfield and Goodhardt (1970), the NBD has the limitation that it may not be appropriate when the product is not sold in packets, since the Poisson part of the model cannot hold. Moreover, for certain heavily bought products, the theoretical variance tends to be systematically higher than the observed variance (Chatfield, 1969). Chatfield and Goodhardt (1970) then proposed the beta-binomial distribution whose results do not suffer from the above limitations of the NBD model, since it is based on the number of purchasing weeks, instead of number of purchases.

Banerjee and Bhattacharya (1976) postulated an inverse Gaussian model for the distribution of time between successive purchases of an individual

household using the bivariate natural conjugate family of distributions for modelling the heterogeneity of the population of households. Specifically, they used the inverse Gaussian distribution (IGD) as a model for the usage time per ounce of toothpaste for an individual household. They also derived the distribution of the number of ounces of toothpaste purchased per household in a time interval of length t .

An IGD with parameters ψ and λ , denoted by $IG(\psi, \lambda)$ has the probability density function (pdf) given by (Banerjee and Bhattacharya, 1976),

$$f(x; \psi, \lambda) = (\lambda/2\pi)^{\frac{1}{2}} x^{-\frac{3}{2}} \exp \left[-\lambda x (\psi - 1/x)^2 / 2 \right], \quad x > 0, \lambda, \psi > 0. \quad (1)$$

The mean and variance of the $IGD(\psi, \lambda)$ distribution are ψ^{-1} and $\psi^{-3}\lambda^{-1}$ respectively. Thus, if the time between successive occurrences of an event follows an $IGD(\psi, \lambda)$ distribution, then the parameter ψ can be interpreted as the average number of occurrences of the event per unit time. For a detailed review of the shapes of the density curve and the properties of the IGD, see, for example, Tweedie (1957), Johnson and Kotz (1970) and Folks and Chhikara (1978). The interest in the IGD arises from the fact that the IGD family accommodates a rich variety of shapes aside from its connection with a Brownian motion process. Specifically, if $X(t)$ is a Brownian motion process with a drift $\mu > 0$, then the random time, when the process first hits a specified barrier $d > 0$, has an IG distribution. If the absorbing barrier d is identified as the initial filled capacity of a unit, and the amount of depletion over time is governed by a Brownian motion with μ representing the mean depletion per unit time, then the usage time of the unit will have an IG distribution. This interpretation of IGD as a first passage time distribution suggests its potential usefulness in modelling life times or usage times as an alternative to other conventional models such as the Weibull, gamma and lognormal.

In this paper, we derive the probability mass function of and other associated functions for the mixture of k inverse Gaussian distributions, where k is assumed to follow the zero-truncated Poisson distribution. This compound distribution is applied in studying aspects of the usage times of a frequently purchased consumer product over different segments of use. The first purchase of the product is regarded as the first segment of use. As the product gains acceptability in the household, subsequent purchases constitute the second, third, etc., segments of use of the product. We are interested in the total usage time over a number of segments of use of the product during a specified period of time t .

2. Results

Let k denote the number of segments of use of the product by a household and X_i the length of the i -th segment. Then the usage time for all segments is denoted by

$$Z = X_1 + X_2 + \dots + X_k \quad (2)$$

where X_i follows a reparameterization of the IGD given by Banerjee and Bhattacharya (1976) with $\mu = \psi^{-1}$ and $\alpha = \lambda$.

Thus,

$$f_x(x) = \left(\frac{\alpha}{2\pi}\right)^{\frac{1}{2}} x^{-\frac{3}{2}} \exp\left[-\alpha(x - \mu)^2 / 2\mu^2 x\right], \quad x > 0; \alpha, \mu > 0. \quad (3)$$

The Laplace transform of $f(x)$ is given as

$$f_x^*(s) = \exp\left\{\frac{\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha}\right)^{\frac{1}{2}}\right]\right\}. \quad (4)$$

The random variable K is assumed to follow the zero-truncated Poisson distribution with probability mass function (pmf)

$$P_K(k, c) = [1 / (1 - e^{-\lambda c})] [(\lambda c)^k e^{-\lambda c} / k!], \quad k = 1, 2, \dots \quad (5)$$

The probability generating function of K is given as

$$g_k(s) = [e^{-\lambda c} / (1 - e^{-\lambda c})] [e^{\lambda cs} - 1] \quad (6)$$

where c is the cut-off time or the period of observation.

The mean number of segments of use at cut-off time c is given as

$$E(K) = \lambda c / (1 - e^{-\lambda c}) \quad (7)$$

and the variance

$$Var(K) = [1 / (1 - e^{-\lambda c})] \{1 - [\lambda c e^{-\lambda c} / (1 - e^{-\lambda c})]\} \quad (8)$$

The maximum likelihood estimator $\hat{\lambda}$ of λ , satisfies the equation

$$\bar{K} = \lambda c / (1 - e^{-\lambda c}) \quad (9)$$

where $\bar{K}$ is the average number of segments of use at cut-off time c .

We shall assume that all segments of use of the product are independently and identically distributed with a common pdf $f(x)$ given in equation (3).

Thus, the Laplace transform of the pmf of all segments, $f_z^*(s)$ is given as

$$\begin{aligned} f_z^*(s) &= g_k(f_x^*(s)) = \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \left[\exp \left[\lambda c \left[\exp \left\{ \frac{\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right] \right] - 1 \right] \\ &= \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \sum_{n=1}^{\infty} \frac{(\lambda c)^n}{n!} \exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \end{aligned} \quad (10)$$

Inverting gives

$$f_z(t) = \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \sum_{n=1}^{\infty} \frac{(\lambda c)^n}{n!} L^{-1} \left[\exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right] \quad (11)$$

The inverse Laplace transform

$$L^{-1} \left[\exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right],$$

is obtained as follows:

Let $Y = \exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\}$, we then have the second order differential equation with variable coefficient

$$\left(1 + \frac{2s\mu^2}{\alpha} \right) Y'' + \frac{\mu^2}{\alpha} Y' - (n\mu)^2 Y = 0 \quad (12)$$

where Y' and Y'' are the first and second derivatives of Y with respect to s , respectively.

Equation (12) can be written as

$$L(t^2 Y) + \frac{2\mu^2}{\alpha} L[t^2 Y' + 2tY] - \frac{\mu^2}{\alpha} L(tY) - (n\mu)^2 L(Y) = 0 \quad (13)$$

i.e.

$$\frac{2\mu^2}{\alpha} t^2 Y' + t^2 Y + \frac{4\mu^2}{\alpha} tY - \frac{\mu^2}{\alpha} tY - (n\mu)^2 Y = 0 \quad (14)$$

which can be written as

$$\frac{dY}{Y} + \left[\frac{\alpha}{2\mu^2} + \frac{3}{2t} - \frac{\alpha n^2}{2t^2} \right] dt = 0. \quad (15)$$

Solving equation (15) gives

$$\ln Y + \frac{\alpha}{2\mu^2} t + \frac{3}{2} \ln t + \frac{\alpha n^2}{2t} = c_1 \quad (16)$$

i.e.

$$Y = \frac{c_1}{t^{\frac{3}{2}}} \exp \left\{ - \left[\frac{\alpha}{2\mu^2} t + \frac{\alpha n^2}{2t} \right] \right\}. \quad (17)$$

Now,

$$tY = \frac{c_1}{t^{\frac{1}{2}}} \exp \left\{ - \left[\frac{\alpha}{2\mu^2} t + \frac{\alpha n^2}{2t} \right] \right\}. \quad (18)$$

For large t , $tY \rightarrow \frac{c_1}{t^{\frac{1}{2}}}$ and $L(tY) \rightarrow \frac{c_1 \sqrt{\pi}}{s^{\frac{1}{2}}}$, so that

$$\begin{aligned} L(tY) &= -\frac{d}{ds} L(Y) = -\frac{d}{ds} \left[\exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right] \\ &= n\mu \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \left[\exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right]. \end{aligned}$$

For small t ,

$$L(tY) = \frac{n\mu}{\left(1 + \frac{2s\mu^2}{\alpha}\right)^{\frac{1}{2}}} = \frac{n\sqrt{\alpha}}{\sqrt{2s^{\frac{1}{2}}}}.$$

Thus,

$$\frac{c_1\sqrt{\pi}}{s^{\frac{1}{2}}} = \frac{n\sqrt{\alpha}}{\sqrt{2s^{\frac{1}{2}}}} \Rightarrow c_1 = \frac{n\sqrt{\alpha}}{\sqrt{2\pi}}.$$

It follows that

$$\begin{aligned} Y &= L^{-1} \left[\exp \left\{ \frac{n\alpha}{\mu} \left[1 - \left(1 + \frac{2s\mu^2}{\alpha} \right)^{\frac{1}{2}} \right] \right\} \right] \\ &= n \left(\frac{\alpha}{2\pi t^3} \right)^{\frac{1}{2}} e^{-\frac{\alpha(t-n\mu)^2}{2\mu^2 t}}. \end{aligned} \quad (19)$$

Thus,

$$f_z(t) = \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \sum_{n=1}^{\infty} \frac{(\lambda c)^n}{(n-1)!} \left(\frac{\alpha}{2\pi t^3} \right)^{\frac{1}{2}} e^{-\frac{\alpha(t-n\mu)^2}{2\mu^2 t}}, \quad 0 < t < \infty. \quad (20)$$

It is pertinent to remark that whereas the IGD defined in equation (2) has a non-monotonic trajectory, the compound distribution tends to have a monotonically decreasing trajectory. This may be due to the monotonically decreasing nature of the mixing distribution (the zero-truncated Poisson).

The mean and variance of the compound distribution are given as

$$E(Z) = (\mu\lambda c) (1 - e^{-\lambda c})^{-1} \quad (21)$$

$$Var(Z) = [\mu^2 \lambda c (\alpha - \mu) + \mu^2 \lambda c (\mu - \alpha - \alpha \lambda c) e^{-\lambda c}] [\alpha (1 - e^{-\lambda c})^2]^{-1} \quad (22)$$

The distribution function is derived as:

$$\begin{aligned} F_z(t) &= \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \sum_{n=1}^{\infty} \frac{(\lambda c)^n}{n!} \left\{ e^{\frac{2n\alpha}{\mu}} - e^{\frac{2n\alpha}{\mu}} \Phi \left[(t + n\mu) \sqrt{\frac{\alpha}{\mu^2 t}} \right] \right. \\ &\quad \left. + \Phi \left[(t - n\mu) \sqrt{\frac{\alpha}{\mu^2 t}} \right] \right\} \end{aligned} \quad (23)$$

where

$$\Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-z^2/2} dz$$

is the standard normal distribution function.

The survival or retention function which gives the probability that the product is still in use weeks after first purchase for all segments of use, is given as

$$G_z(t) = 1 - F_z(t). \quad (24)$$

The "half-life" $t_{0.5}$ which is the number of weeks since first purchase during which there is a 50% chance that the product is still in use, is obtained from the equation

$$G_z(t_{0.5}) = 0.5. \quad (25)$$

The hazard function (termination rate) is given as

$$m_z(t) = \frac{f_z(t)}{G_z(t)}. \quad (26)$$

The average duration of use for all segments of use of the product t weeks after first purchase is given as

$$\begin{aligned} D_z(t) &= \int_0^t G_z(r) dr \\ &= t - \frac{e^{-\lambda c}}{1 - e^{-\lambda c}} \sum_{n=1}^{\infty} \frac{(\lambda c)^n}{n!} \left\{ \theta t + \frac{(2n\alpha\mu\theta + \mu^2)}{2\alpha} - [(t + n\mu)\theta] \Phi_1(t) \right. \\ &\quad \left. + \left[\frac{(\alpha t - \mu^2 - n\alpha\mu)}{\alpha} \right] \Phi_2(t) + \frac{\mu^2}{\alpha\sqrt{2\pi}} \alpha(t) \beta(t) \right\} \end{aligned} \quad (27)$$

where

$$\begin{aligned} \theta &= e^{\frac{2n\alpha}{\mu}}, \quad \Phi_1(t) = \Phi\left[(t + n\mu)\sqrt{\frac{\alpha}{\mu^2 t}}\right], \quad \Phi_2(t) = \Phi\left[(t - n\mu)\sqrt{\frac{\alpha}{\mu^2 t}}\right] \\ \alpha(t) &= \left[(t - n\mu)\sqrt{\frac{\alpha}{\mu^2 t}}\right], \quad \beta(t) = e^{-\frac{\alpha(t - n\mu)^2}{2\mu^2 t}}. \end{aligned}$$

The mean duration of use for all segments of use from the first purchase to the cut-off time c is given as

$$D_z(c) = D_z(t) \quad \text{at } t = c. \quad (28)$$

3. Application

We illustrate the results derived using the results derived using the data for the interpurchase times per ounce of toothpaste for individual households. The parameter estimates for the IGD for four households are given as follows (Banerjee and Bhattacharya, 1976):

Household I:	$\mu = 0.4219,$	$\alpha = 1.24$
Household II:	$\mu = 0.2340,$	$\alpha = 0.791$
Household III:	$\mu = 0.8658$	$\alpha = 1.492$
Household IV	$\mu = 0.5420$	$\alpha = 1.152$

The distribution of the number of purchases for all segments of use for a cut-off time $c = 5$ weeks is presented below (Banerjee and Bhattacharya, 1976):

No. of purchases (k)	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Frequency	57	63	55	36	17	15	12	10	1	3	3	0	0	1

The mean number of purchases $\bar{K} = 3.3589$. The maximum likelihood estimate of the parameter λ is obtained from the equation

$$3.3589 = \frac{5\hat{\lambda}}{1 - e^{-5\hat{\lambda}}} \quad (29)$$

Thus $\hat{\lambda} = 0.645$.

The computed values of the pmf $f_z(t)$ of the usage time for all segments of use for each of the four households, the survival function $G_z(t)$, the length of use attrition or termination rate $m_z(t)$, the average duration of use t weeks after first purchase $D_z(t)$ (hence $D_z(5)$) and the "half-life" $t_{0.5}$ are presented in the table below.

Table showing $f_z(t)$, $G_z(t)$, $m_z(t)$, $D_z(t)$ and $t_{0.5}$ for each of four households

t	Household I				Household II			
	$f_z(t)$	$G_z(t)$	$m_z(t)$	$D_z(t)$	$f_z(t)$	$G_z(t)$	$m_z(t)$	$D_z(t)$
1	0.503	0.67	0.751	0.845	0.627	0.308	2.036	0.709
2	0.286	0.247	1.158	1.310	0.055	0.019	2.895	0.838
3	0.083	0.060	1.383	1.470	0.0016	0.0005	3.200	0.884
4	0.017	0.011	1.545	1.528	0	0	∞	0.884
5	0.0025	0.0015	1.676	1.577	0	0	∞	0.884
$t_{0.5}$	1.351				0.748			
T	Household III				Household IV			
	$f_z(t)$	$G_z(t)$	$m_z(t)$	$D_z(t)$	$f_z(t)$	$G_z(t)$	$m_z(t)$	$D_z(t)$
1	0.215	0.888	0.242	0.793	0.380	0.771	0.493	0.845
2	0.236	0.675	0.350	1.631	0.302	0.415	0.728	1.456
3	0.193	0.458	0.421	2.214	0.151	0.175	0.863	1.750
4	0.134	0.285	0.470	2.586	0.058	0.061	0.951	1.889
5	0.083	0.166	0.500	2.822	0.019	0.019	1	1.970
$t_{0.5}$	2.796				1.744			

Graphs of $f_z(t)$ against values of t for each of the four households are presented in figure 1, and those for $m_z(t)$ against values of t are presented in figure 2.

The table as well as the graphs show that the rate of termination after one week of use is 0.751 for household I and 2.036 for household II. This rate after five weeks of use is 0.5 and 1 for households III and IV respectively. The median duration of use (i.e. the number of weeks of use such that the probability of termination is 0.50) are 1.35, 0.75, 2.80 and 1.74 for households I, II, III and IV respectively.

The mean duration of use five weeks after first purchase are 1.58 weeks, 0.88 weeks, 2.82 weeks and 1.97 weeks for household I, II, III and IV, respectively.

The survival functions show that the probability of continuous use of the product is 0.06 after 3 weeks for household I, 0.02 after 2 weeks for household

II, 0.17 after 5 weeks for household III and 0.06 after 4 weeks for household IV. Thus, household I is likely to use the product for up to two weeks, household II for up to one week, household III for slightly more than five weeks and household IV for up to three weeks. It is therefore clear from the results that brand loyalty is strongest for household III, followed by household IV and household I. It is, however weak for household II.

4. Conclusion

We have derived the probability distribution of and other related functions for the mixture of the zero truncated Poisson and the inverse Gaussian distributions with the former as the mixing distribution. This distribution is likely to find application in modelling aspects of the duration of use of consumer goods for all segments of use, the length of life of equipment and components for all periods of use, the length of use of a contraceptive device for all segments of use, etc. In this paper, the distribution is applied in the context of usage of consumer goods and the results were found to be realistic.

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